Algorithm 1 Iterative Fine-tuning of a Generative Model With Correction

Input: $\mathcal{D}_{\text{real}} := \{x_i\}_{i=1}^n$, $\mathcal{A}$, $\mathcal{A}_{\text{ft}}$, π_γ // ground truth data, learning procedure, fine-tuning procedure, correction function
Parameters: T, λ, γ // number of retraining iterations, proportion of generated data, correction strength
$p_{\theta_0} \leftarrow \mathcal{A}(\mathcal{D}_{\text{real}})$ // learn generative model from scratch on true data
for $t = 1$ **to** T **do**
  $\mathcal{D}_{\text{synth}} \leftarrow \{\pi_\gamma(\tilde{x}_i)\}_{i=1}^{\lfloor \lambda \cdot n \rfloor}$, with $\tilde{x}_i \sim p_{\theta_{t-1}}$ // sample $\lfloor \lambda \cdot n \rfloor$ synthetic data points, pass through correction function
  $p_{\theta_t} \leftarrow \mathcal{A}_{\text{ft}}(\mathcal{D}_{\text{real}} \cup \mathcal{D}_{\text{synth}}; p_{\theta_{t-1}})$ // fine-tune previous generation using augmented dataset
end for
Return $[p_{\theta_0}, p_{\theta_1}, p_{\theta_2}, \ldots, p_{\theta_T}]$

model in our generative model class, this means that as γ increases from 0 to ∞, we have that $\pi_\gamma p_\theta(x)$ is a PDF which better represents the target likelihood that we want to estimate through training the generative model.

In our theoretical formulation, we consider correction functions that correct the probability distribution p_θ, rather than the more intuitive (and practical) case of a correction function that corrects individual points that the distribution is defined over. In Appendix C, we specify sufficient conditions under which a pointwise correction function is guaranteed to correspond to a distribution-wise correction function of the same form as those which we consider in our theoretical study and therefore can enjoy the theoretical stability guarantees we prove. We also provide a concrete example of a projection function, in the Gaussian case, which provably satisfies those conditions. We conduct a series of experiments on this toy example in Section 5.

4.1.2. Understanding the weight update $\pi_\gamma \mathcal{G}_\lambda^n(\theta)$

The weight update $\pi_\gamma G_\lambda^n(\theta)$ in (5) is a formalization of the intended output of fine-tuning p_θ on $\mathcal{D}_{\text{real}} \cup \mathcal{D}_{\text{synth}}$, where $\mathcal{D}_{\text{real}} = \{x_i\}_{i=1}^n$ is the ground truth dataset of size n, and $\mathcal{D}_{\text{synth}} = \{\tilde{x}_i : \tilde{x}_i \sim \widehat{\pi_\gamma p_\theta}\}_{i=1}^{\lfloor \lambda \cdot n \rfloor}$ is the synthesized-and-corrected dataset of size $\lfloor \lambda \cdot n \rfloor$. In other words, in an ideal run of stochastic gradient descent fine-tuning, the model weights θ should update to $\pi_\gamma \mathcal{G}_\lambda^n(\theta)$, as defined in (5), when trained on $\mathcal{D}_{\text{real}} \cup \mathcal{D}_{\text{synth}}$.

Intuitively, the weight update $\theta \mapsto \pi_\gamma \mathcal{G}_\lambda^n(\theta)$ avoids the loss of variance in the generated data by ensuring that at each step, the model is trained on synthetic data which is likelier to have been sampled from the diverse target distribution. This positive phenomenon is more pronounced when the correction strength γ is larger.

4.2. Assumptions

In order to prove our main result, we need some regularity assumptions about the learning procedure. Informally speaking, we will assume that the class of generative models that we consider is smoothly parameterized by its model weights; the loss landscape is concave near the ideal model weights; and the class of generative models does an increasingly good job approximating the target data distribution as the dataset size increases. We formally quantify and state these hypotheses in Assumption 4.2.

Assumption 4.2. The following are true.

1. There exists some $L > 0$ such that, for all θ sufficiently close to $\theta^\star$, the mapping $x \mapsto \nabla_\theta^2 \log p_\theta(x)$ is L-Lipschitz.
2. The mapping $\theta \mapsto \mathbb{E}_{x \sim p_{\text{data}}}[\log p_\theta(x)]$ is continuously twice differentiable locally around $\theta^\star$, and there exists some $\alpha > 0$ such that $\mathbb{E}_{x \sim p_{\text{data}}}\left[\nabla_\theta^2 \log p_\theta(x)\right]|_{\theta^\star} \preceq -\alpha I_d \prec 0$.
3. There exist $a, b, \varepsilon_{\text{OPT}} \geq 0$ and a neighborhood U of $\theta^\star$ such that, for any $\delta \in (0,1)$, with probability $1-\delta$ over the samplings, we have[2]
$$\|\pi_\gamma \mathcal{G}_\lambda^n(\theta) - \pi_\gamma \mathcal{G}_\lambda^\infty(\theta)\| \leq \varepsilon_{\text{OPT}} + \frac{a}{\sqrt{n}} \sqrt{\log \frac{b}{\delta}}. \quad (6)$$
for all $\theta \in U$ and $n \in \mathbb{N}$. Denote this bound by $\tau_n(\delta)$.

In Assumption 4.2 (2), the notation "$\preceq$" corresponds to the Loewner order on symmetric matrices: we write that $A \preceq B$ if $B - A$ is positive semi-definite, and $A \prec B$ if $B - A$ is positive definite. In particular, Assumption 4.2 (2) implies that the matrix $\mathbb{E}_{x \sim p_{\text{data}}}\left[\nabla_\theta^2 \log p_\theta(x)\right]|_{\theta^\star}$ is negative definite, and its largest eigenvalue is at most $-\alpha$. And Assumption 4.2 (3) mirrors the main assumption in (Bertrand et al., 2024); it is motivated by generalization bounds in deep learning, see e.g. (Jakubovitz et al., 2019; Ji et al., 2021). The interested reader can consult Appendix B for more details on this assumption.

4.3. Iterative Fine-Tuning with Correction

We now have the language to state our main result, which essentially says that if the initial parameters θ_0 are suffi-

[2]The map $\pi_\gamma G_\lambda^\infty$ is defined similarly to $\pi_\gamma G_\lambda^n$ in (5), but with $\hat{p}_{\text{data}}$ replaced with p_{data}, and with $\widehat{\pi_\gamma p_\theta}$ replaced with $\pi_\gamma p_\theta$. See Appendix A for more details. This estimate is identical to the analogous Assumption 3 used in (Bertrand et al., 2024), with the only difference being it is applied to our iterative fine-tuning update function. See Appendix B for further discussion.

ciently close to the optimal model parameters $\theta^\star$, and if the augmentation percentage λ is sufficiently small, then under iterative fine-tuning with correction, we can expect our subsequent model parameters to stay close to $\theta^\star$.

Theorem 4.3 (Stability of Iterative Fine-Tuning with Correction)**.** *Fix an augmentation percentage $\lambda \in \mathbb{R}_{>0}$ and a correction strength $\gamma \in \mathbb{R}_{\geq 0}$. Suppose we have an iterative fine-tuning procedure defined by the rule $\theta^n_{t+1} = \pi_\gamma \mathcal{G}^n_\lambda(\theta^n_t)$, and suppose that Assumption 4.2 holds. Define the constant*

$$\rho(\lambda) := \rho(\lambda; \alpha, \varepsilon, L) := \frac{\lambda(\alpha + \varepsilon L)}{\alpha - \lambda(\alpha + \varepsilon L)}$$

and fix any $\delta \in (0, 1)$. If θ_0 is sufficiently close to $\theta^\star$, and if $\lambda\left(1 + \frac{\varepsilon L}{\alpha}\right) < \frac{1+\gamma}{2+\gamma}$, then $\rho(\lambda)/(1+\gamma) < 1$, and it follows that the stability estimate holds with probability $1 - \delta$:

$$\|\theta^n_t - \theta^\star\| \leq \tau_n(\delta/t) \sum_{i=0}^{t} \left(\frac{\rho(\lambda)}{1+\gamma}\right)^i + \left(\frac{\rho(\lambda)}{1+\gamma}\right)^t \|\theta^n_0 - \theta^\star\| \tag{7}$$

for all $t > 0$.

We prove Theorem 4.3 in Appendix A.

Remark 4.4. If we apply Theorem 4.3 with correction strength $\gamma = 0$, then the iterative fine-tuning procedure trains successively on a combination of raw synthetic data that has not been corrected using a correction function and ground truth data. This is exactly the case considered in (Bertrand et al., 2024). Accordingly, the bound in (7), applied with $\gamma = 0$, exactly recovers their result.

Corollary 4.5. *Under the assumptions from Theorem 4.3, iterative fine-tuning with any amount of correction outperforms iterative fine-tuning without correction–in the sense that it is exponentially more stable, and it results in better model weights.*

Proof of Corollary 4.5. We apply Theorem 4.3 with $\gamma = 0$, which corresponds to no correction, as well as with $\gamma > 0$, which corresponds to any amount of correction. For any $\gamma > 0$, we notice that the RHS of (7) is strictly smaller than when $\gamma = 0$. This guarantees better stability as $t \to \infty$, as well as model weights θ^n_t closer to $\theta^\star$. □

Example 4.6. If we apply Theorem 4.3 with correction strength $\gamma \to \infty$, then the bound (7) in Theorem 4.3 limits to $\tau_n(\delta/t)$. This implies that the practical iterate θ^n_t approaches the ideal model paramaters, and is at worst some constant away, that depends on error from the optimization procedure, as well as statistical error from using finitely many ground truth data samples n.

Note that Theorem 4.3 relies on the assumption that the initial model parameters θ_0 are sufficiently close to the ideal model parameters $\theta^\star$, and also that the augmentation percentage λ is sufficiently small. We hypothesize that these assumptions can be relaxed in the case where a correction function participates in the iterative fine-tuning procedure–intuitively, the correction function should compensate for errors that arise from θ^n_0 being worse, as well as errors that arise from incorporating more synthetic data. We frame this in the following conjecture.

Conjecture 4.7. *In the case of iterative fine-tuning with correction, we may relax how close the initial model parameters θ^n_0 need to be to the optimal model parameters $\theta^\star$, as well as choose a larger synthetic augmentation percentage λ, while still retaining the improved stability estimate (7).*

We provide empirical evidence for Conjecture 4.7 in Section 7 on the human motion synthesis task. In fact, Theorem 4.3 represents partial progress towards this conjecture. Namely, according to Theorem 4.3, for large correction strength γ, we can effectively choose a synthetic augmentation percentage that is twice as large as we would be able to without any correction, and still be able to meet the assumptions of the theorem. This is because $\lim_{\gamma\to\infty} \frac{1+\gamma}{2+\gamma} = 1$, which is twice as large as the bound when $\gamma = 0$.

5. Toy Example: Gaussian

We first assume oracle knowledge of the ground truth distribution, and use a toy example to directly demonstrate the impact of the correction strength γ on model performance and stability as stated in Theorem 4.3 and Corollary 4.5. Our ground truth distribution is a 2-dimensional isotropic Gaussian centered at the origin, i.e., $\theta^\star = ((0, 0), I_2)$, and our correction is "distribution-wise" in this idealized scenario. We consider the more practical setting, where we don't have oracle knowledge of the target distribution a priori, and where the data correction is "point-wise", in the empirical studies in the following two sections. Further, in Appendix C, we show that, in theory, sufficiently well-behaved pointwise correction functions indeed correspond to distribution-wise correction functions.

Concretely, our ground truth dataset contains 50 points sampled from the target distribution, which are used to estimate $\theta^{50}_0 = (\mu_0, \Sigma_0) \in \mathbb{R}^6$. We fix our synthetic augmentation percentage to be $\lambda = 0.5$, and inductively synthesize a new dataset $\mathcal{D}_{\text{synth}} = \{y_i \sim \mathcal{N}(\mu_t, \Sigma_t)\}_{i=1}^{25}$. We implement a correction function to map $\mathcal{D}_{\text{synth}}$, which was sampled from $p_{\theta^{50}_t}$, to a dataset $\mathcal{D}_{\text{corrected}}$, which is likelier to have been sampled from the target density $p_{\theta^\star}$. We do this by sampling $\mathcal{D}_{\text{corrected}}$ from the *middle density* corresponding to a given correction strength γ:

$$\pi_\gamma \hat{p}_{\theta^{50}_t}(x) := \frac{\hat{p}_{\theta^{50}_t}(x) + \gamma p_{\theta^\star}(x)}{1+\gamma}, \tag{8}$$

where $\hat{p}_{\theta^{50}_t}$ is the empirical PDF obtained from $\mathcal{D}_{\text{synth}}$.

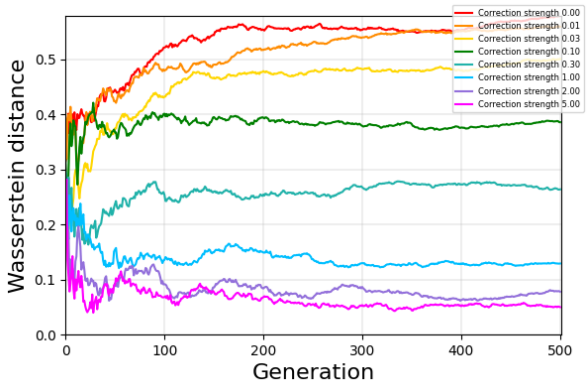


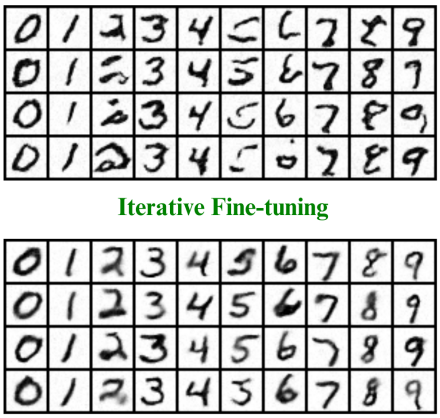
Figure 2. Empirical results from our Gaussian toy example. The graph demonstrates that increasing the correction strength γ, with a fixed augmentation ratio of $\lambda = 0.5$, improves performance and stability after self-consuming iterations.

We logarithmically accrue synthetic data points to simulate the case of fine-tuning. We obtain the updated model parameters θ_{t+1}^{50} by computing the sample mean and covariance on this augmented dataset. In Figure 2, we present the Wasserstein distance between the origin-centered isotropic Gaussian target distribution and the distribution defined by the parameters θ_t^{50} at each iteration t. Our results illustrate how increasing the correction strength γ adds stability and results in convergence near better Wasserstein scores in later generations, in accordance with Theorem 4.3. The experiments also demonstrate how even a very small increase in γ can improve performance over the baseline, in accordance with our claim of exponential improvement in Corollary 4.5.

6. Toy Example: MNIST

Our proof uses the optimal target PDF $p_{\theta^\star}$ to define the correction function π_γ. This is empirically validated by the Gaussian toy experiment, which assumes knowing the true target distribution. In practice, the correction function only depends on the ability to map synthesized data to data which is *likelier* to have been sampled from the ground truth distribution. Crucially, *this can be achieved without having a complete description of the target distribution.* For example, with our human motion experiments, we will demonstrate that point-wise correction based on the laws of physics is one proxy approach to make a sample more likely, without knowing the true target distribution.

One has the freedom to explore alternative approaches to data correction for more general data types, such as images. For example, one simple heuristic is to identify the "anchor" or "exemplar" images, which are intuitively representative and likely. The correction function can then be implemented as mapping or morphing synthesized data towards its nearest anchor, to make the synthesized data more representative and likely. In this section, we implement this approach on MNIST and study its performance.



Iterative Fine-tuning

Iterative Fine-tuning with Self-Correction

Figure 3. Empirical results from our MNIST toy example. These synthesized images demonstrate that after 50 self-consuming iterations at 150% augmentation percentage, the model which is trained using **iterative fine-tuning with self-correction** is able to generate higher quality samples than the model trained using **iterative fine-tuning** without any self-correction.

For our MNIST (LeCun et al., 1998) experiments, we train a diffusion model (Ho et al., 2020) for class-conditional image generation, using a train split of size $n = 12000$. For our iterative fine-tuning experiments, we train the model for 20 epochs, then synthesize $\lambda \cdot 12000/10$ images for each digit, and then augment the ground truth dataset with these to train on for the next generation; every following generation follows the same procedure, but only trains for a single epoch. We vary our experiments over augmentation percentages $\lambda \in \{0.2, 0.5, 1.0, 1.5\}$. To define our self-correction operation, we first compute K-means clusters over the training split for each digit. Our iterative fine-tuning with self-correction experiments use the same setup described above, except instead of training on the synthesized images, we train on the synthesized and then corrected images, where "correcting" an image means finding the nearest centroid in the K centroids for that digit that we computed at the start of training. We swept the values $K \in \{1, 2, 4, \ldots, 1024\}$, and we found that any reasonably large K results in the same general trend where self-correction improves the metrics and stability. We report our results for $K = 16$, which performs the best.

We present images synthesized using our trained models in Figure 3. These synthesized images demonstrate that iterative fine-tuning eventually generates many low quality and illegible digits, and this problem is solved by applying our self-correction operation. Further experiment details, including graphs of the FID metrics for each generation that provide rigorous evidence for this trend across augmentation percentages, can be found in Appendix D. Our empirical results demonstrate that applying self-correction improves performance during iterative fine-tuning for our MNIST image generation task across self-consuming generations, and